🛡️ Module 07: Perimeter Defence Evasion Techniques

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# Exercise 04: Evading Perimeter Defences Using Social-Engineer Toolkit (SET)
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Scenario

Host-based anti-malware protections (such as Windows Defender) actively monitor process memory and disk write operations to detect known signature patterns and malicious executables. Client-side evasion tactics leverage native script interpreters—such as PowerShell—to dynamically execute shellcode directly within host memory without creating persistent files on disk.

The **Social-Engineer Toolkit (SET)** provides specialized PowerShell Attack Vectors designed to bypass host endpoint protections. By accessing SET module menu option **1** (*Social-Engineering Attacks*) and selecting menu option **9** (**9**, *PowerShell Attack Vectors*), security testers can generate obfuscated PowerShell shellcode payloads (**x86_powershell_injection.txt**). When executed on a target host like Windows Server 2019, the payload establishes an outbound encrypted Meterpreter session back to the handler listening on port **443**.

In this exercise, we initialize SET on Parrot Security, generate a PowerShell injection payload using menu option **9**, start an automated Metasploit payload handler on port **443**, transfer and execute the script in PowerShell on Windows Server 2019, and establish an active Meterpreter shell.

Objective

The objective of this lab is to help students learn how to:
- Launch and configure the **Social-Engineer Toolkit (SET)** (**sudo setoolkit**)
- Navigate SET attack vector menus to locate PowerShell Attack Vectors (Option **1** $
ightarrow$ Option **9**)
- Generate memory-injection PowerShell payloads targeting Windows host platforms
- Configure custom reverse TCP payload handlers listening on port **443**
- Locate generated SET report files in hidden directory paths
(**/root/.set/reports/powershell/x86_powershell_injection.txt**)
- Execute PowerShell payloads on Windows Server 2019 to obtain interactive **Meterpreter** sessions

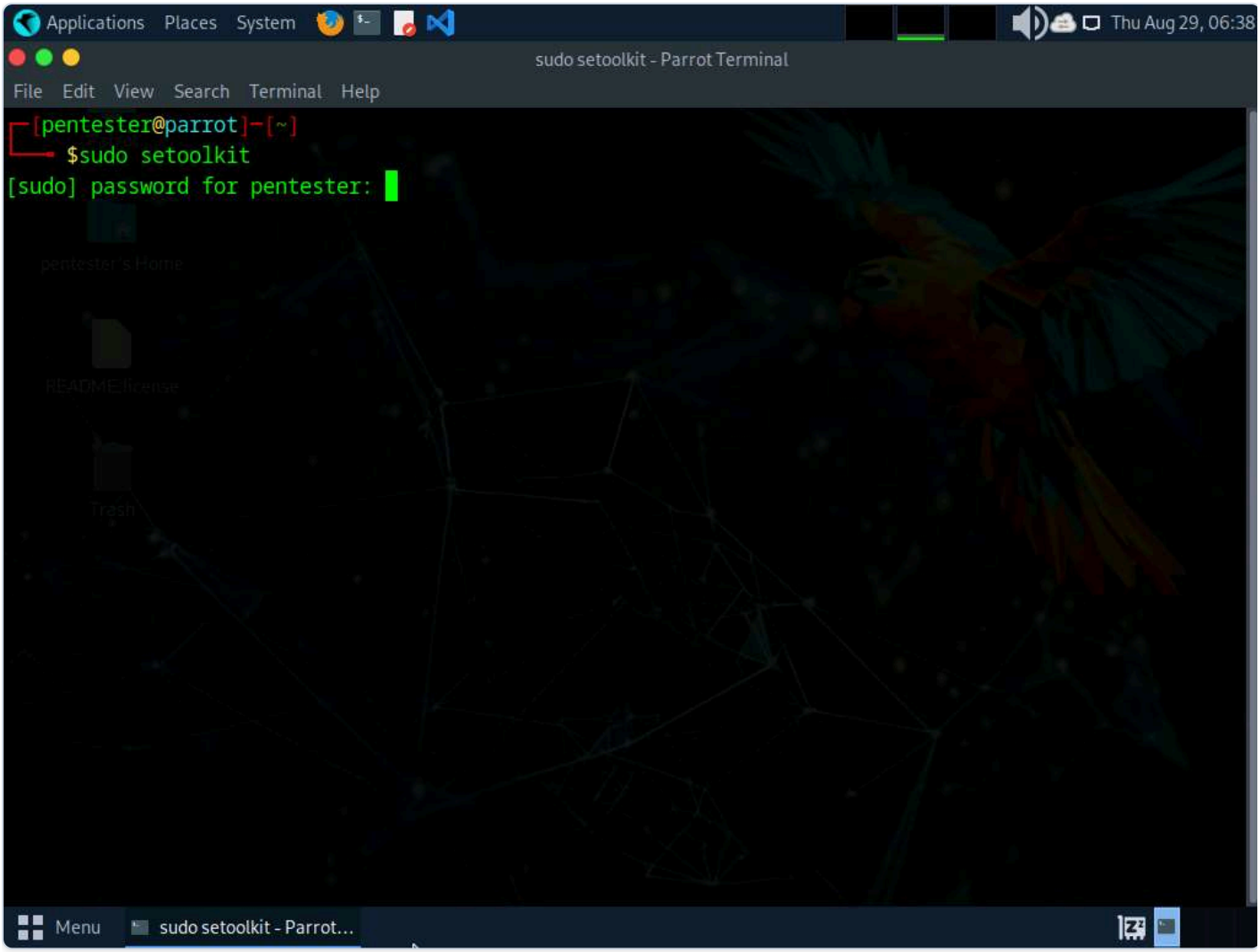
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Task 1: Generate PowerShell Payload using SET and Execute Exploit

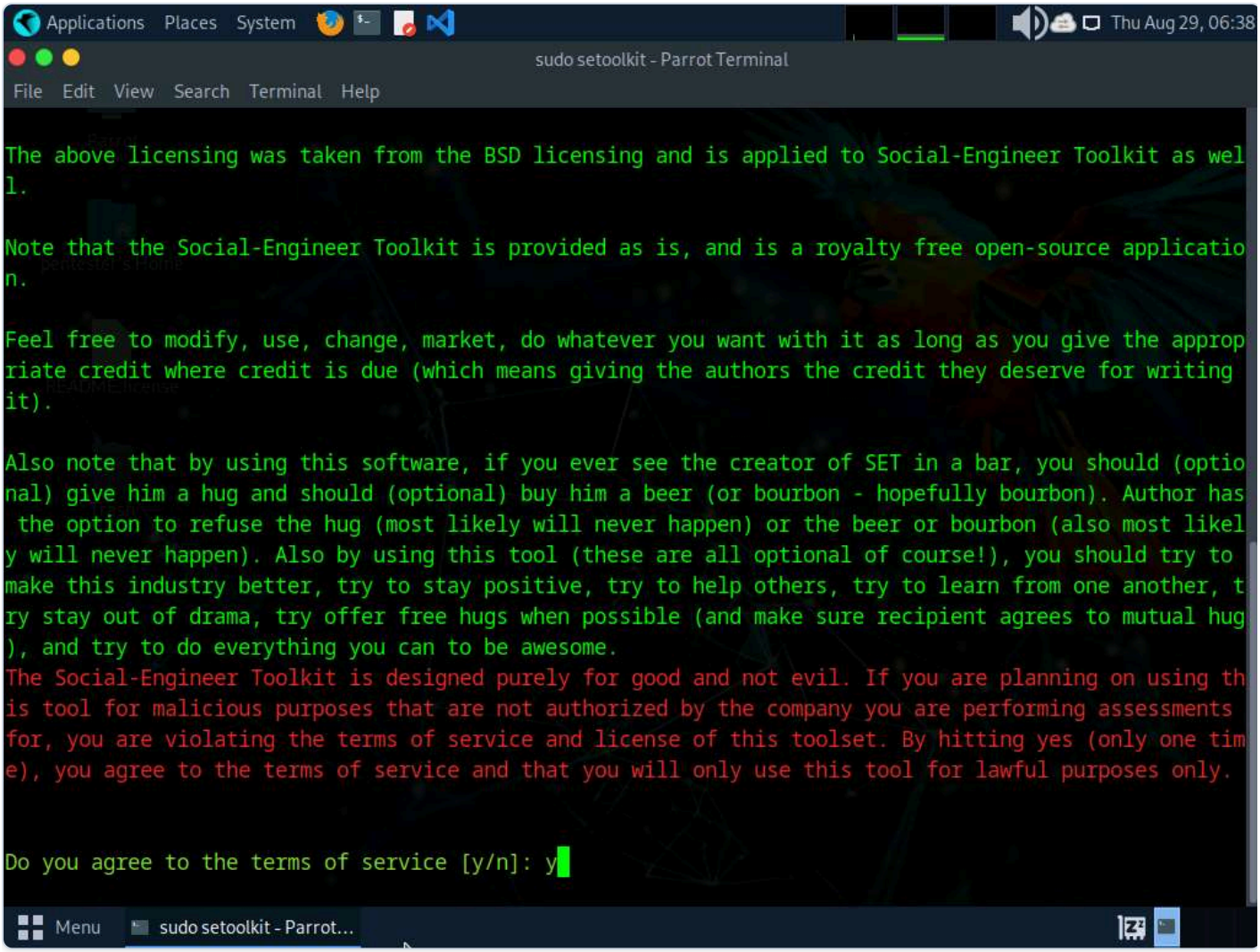
In this task, we launch SET on Parrot Security, generate a PowerShell attack script using option **1** and option **9**, configure the listener for IP **192.168.0.18** on port **443**, copy the payload to Windows Server 2019, and execute it to receive a Meterpreter reverse shell.

Lab Tasks

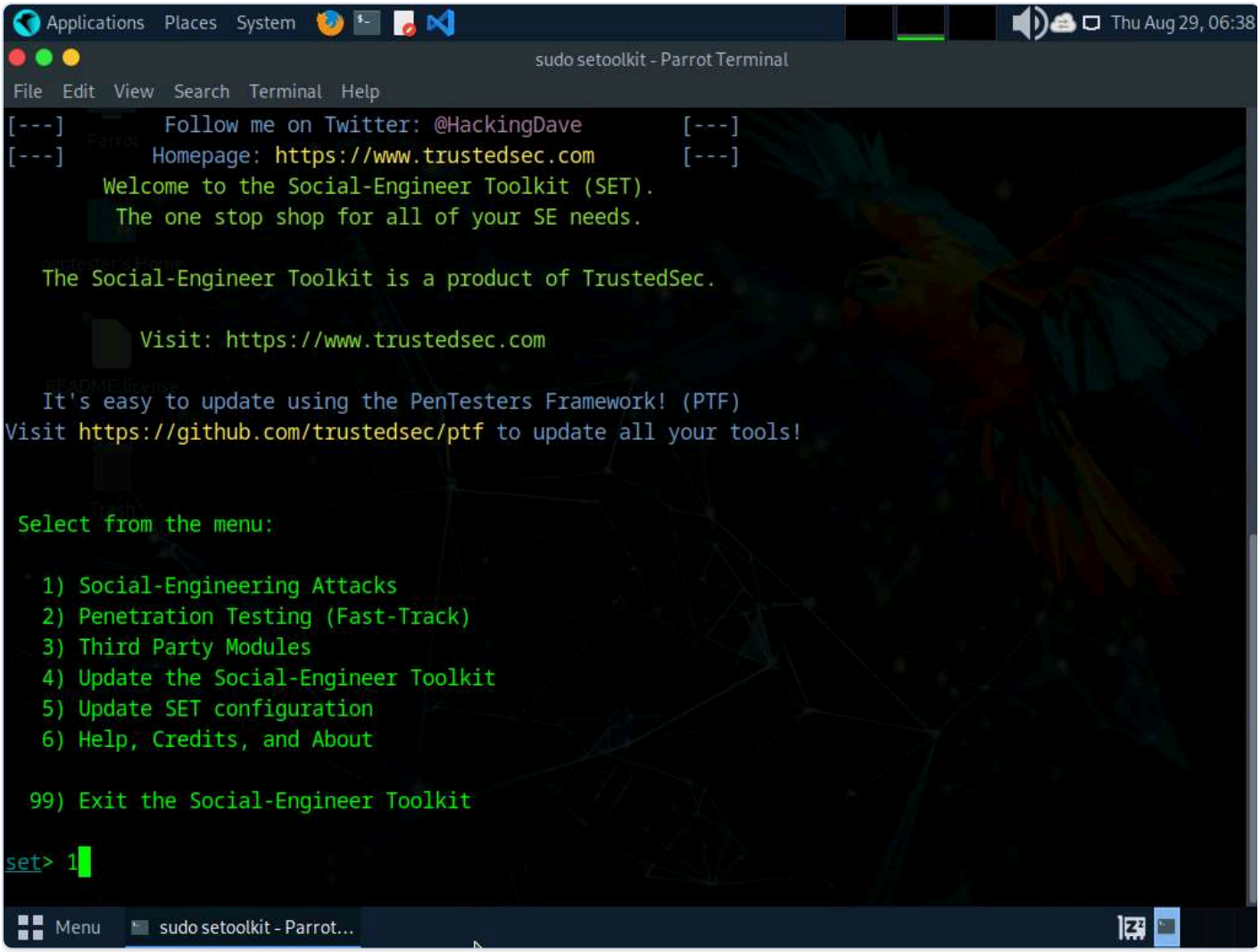
1. Switch to **Parrot Security** (**toor**). Open a terminal and launch the Social-Engineer Toolkit:
   sudo setoolkit



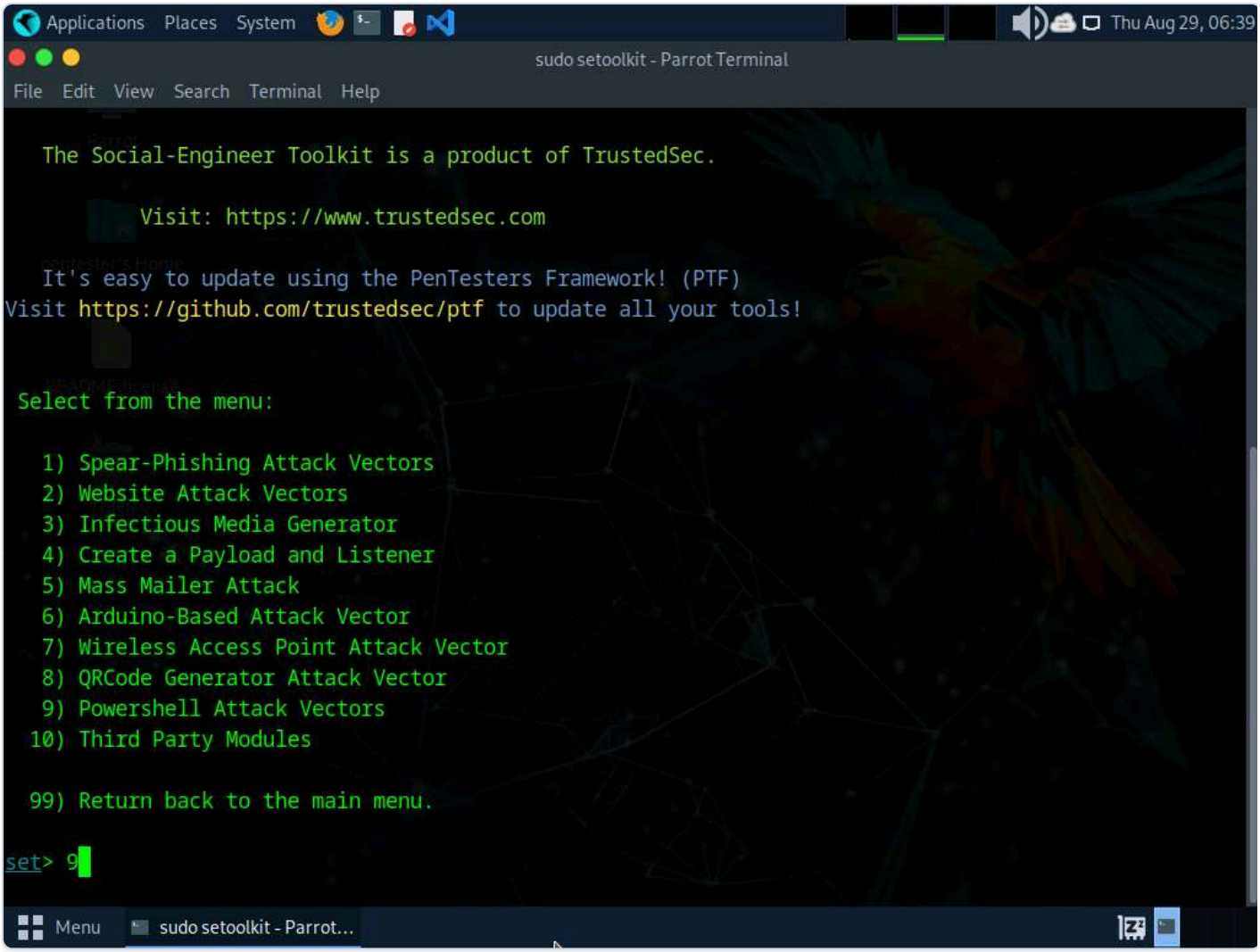
1. Type **y** and press Enter to accept the terms of service.



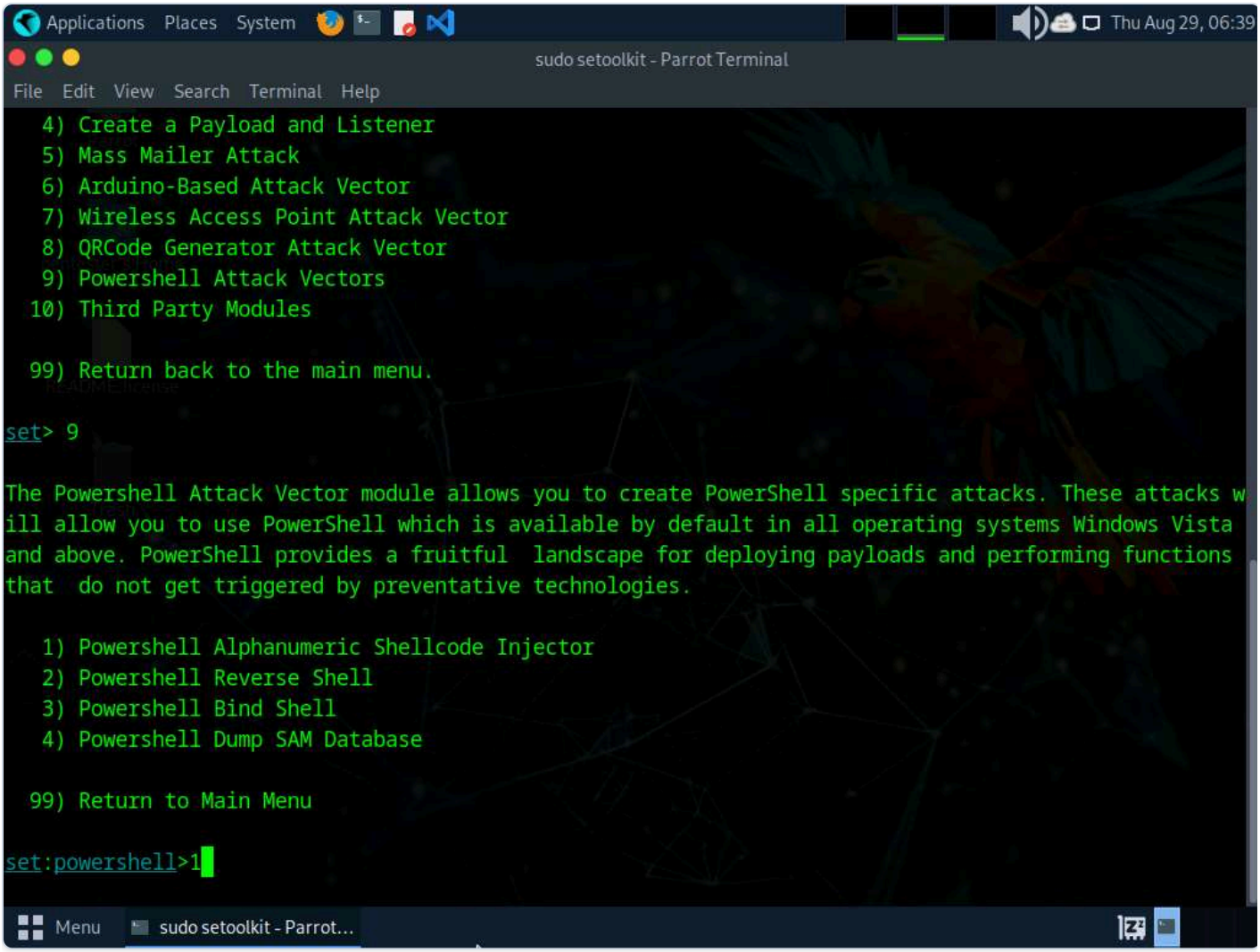
1. Type **1** and press Enter to select **Social-Engineering Attacks**.



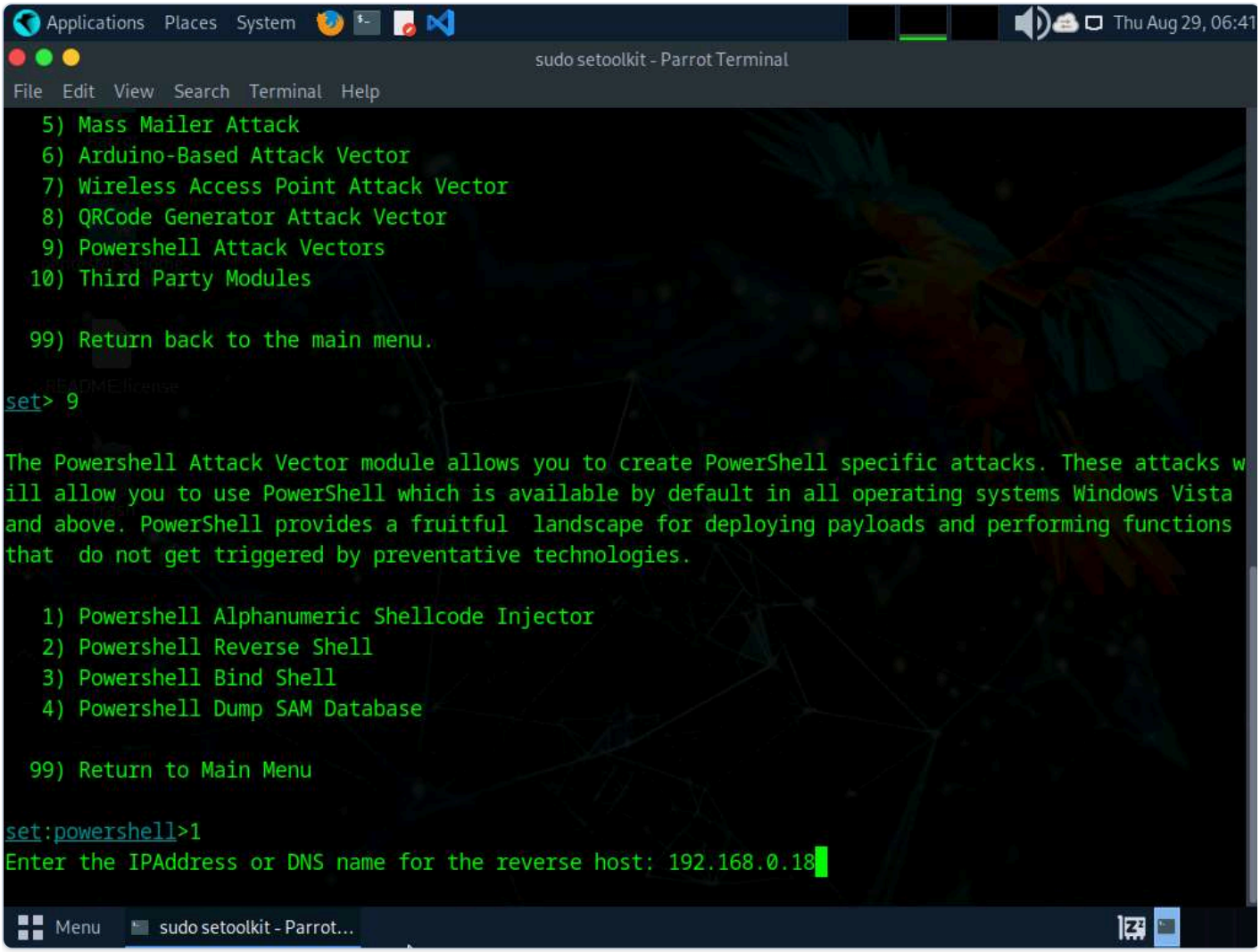
1. Type **9** and press Enter to select **PowerShell Attack Vectors**.



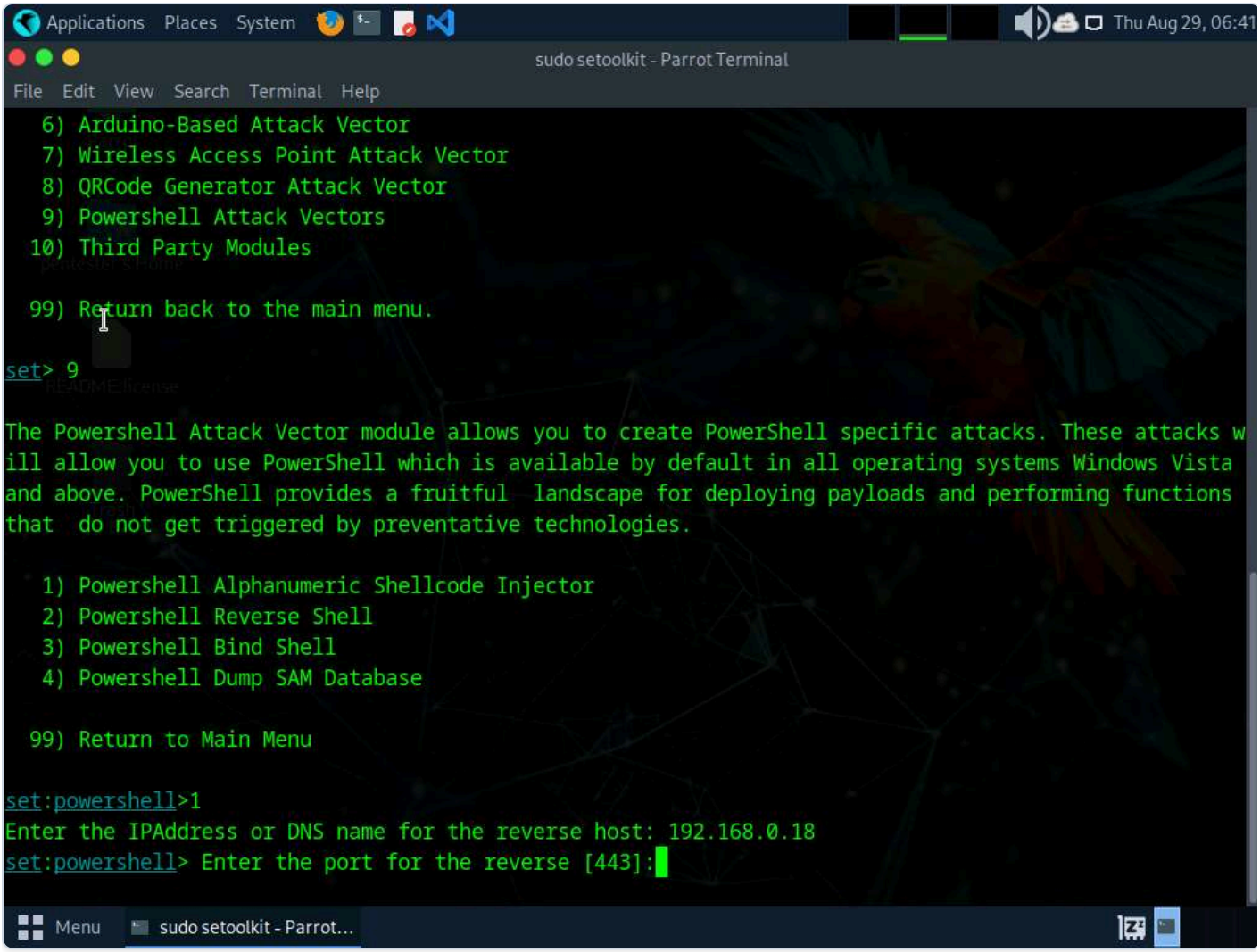
1. Type **1** and press Enter to select **PowerShell Alphanumeric Shellcode Injector**.



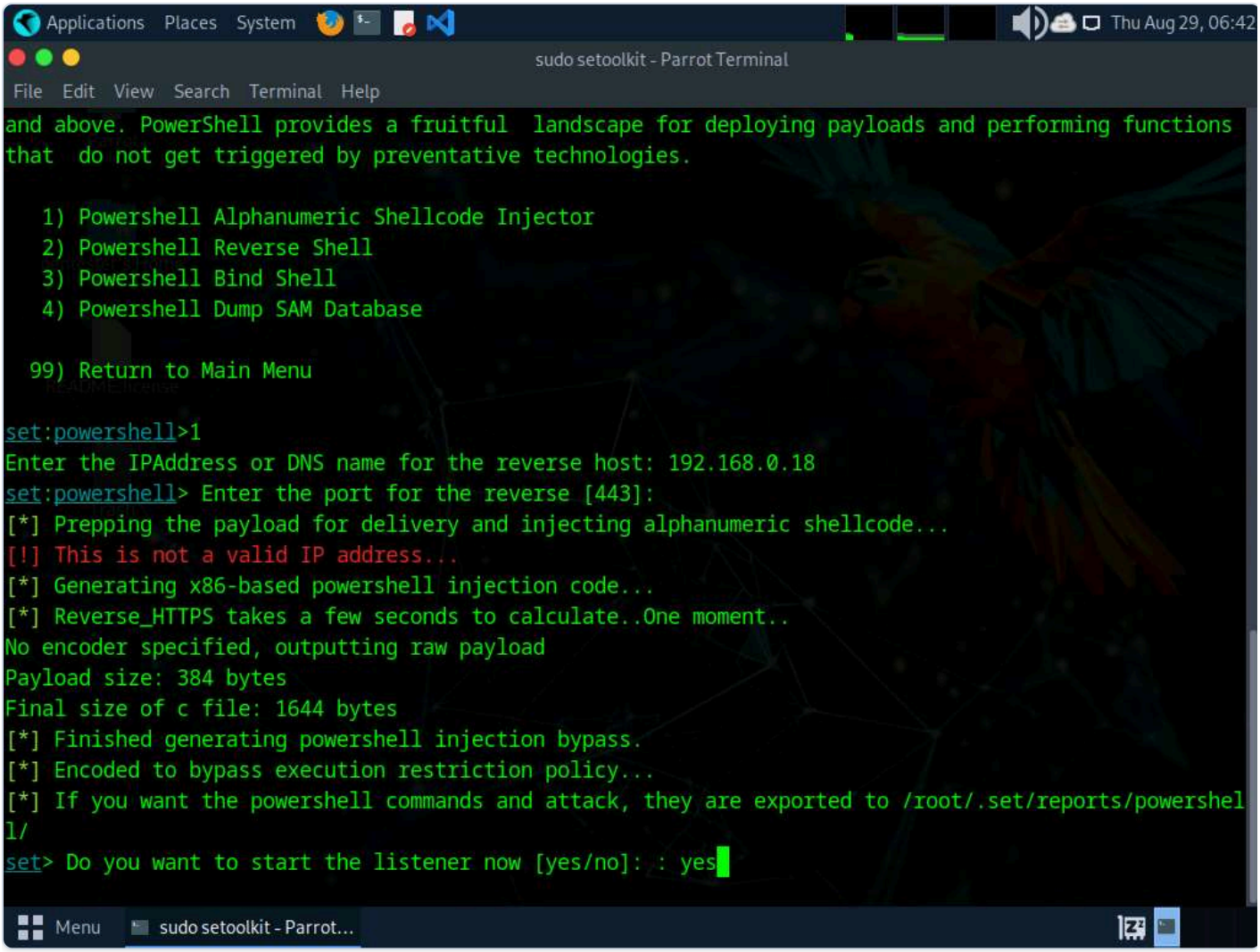
1. Enter the IP address of your Parrot Security handler machine:
   192.168.0.18



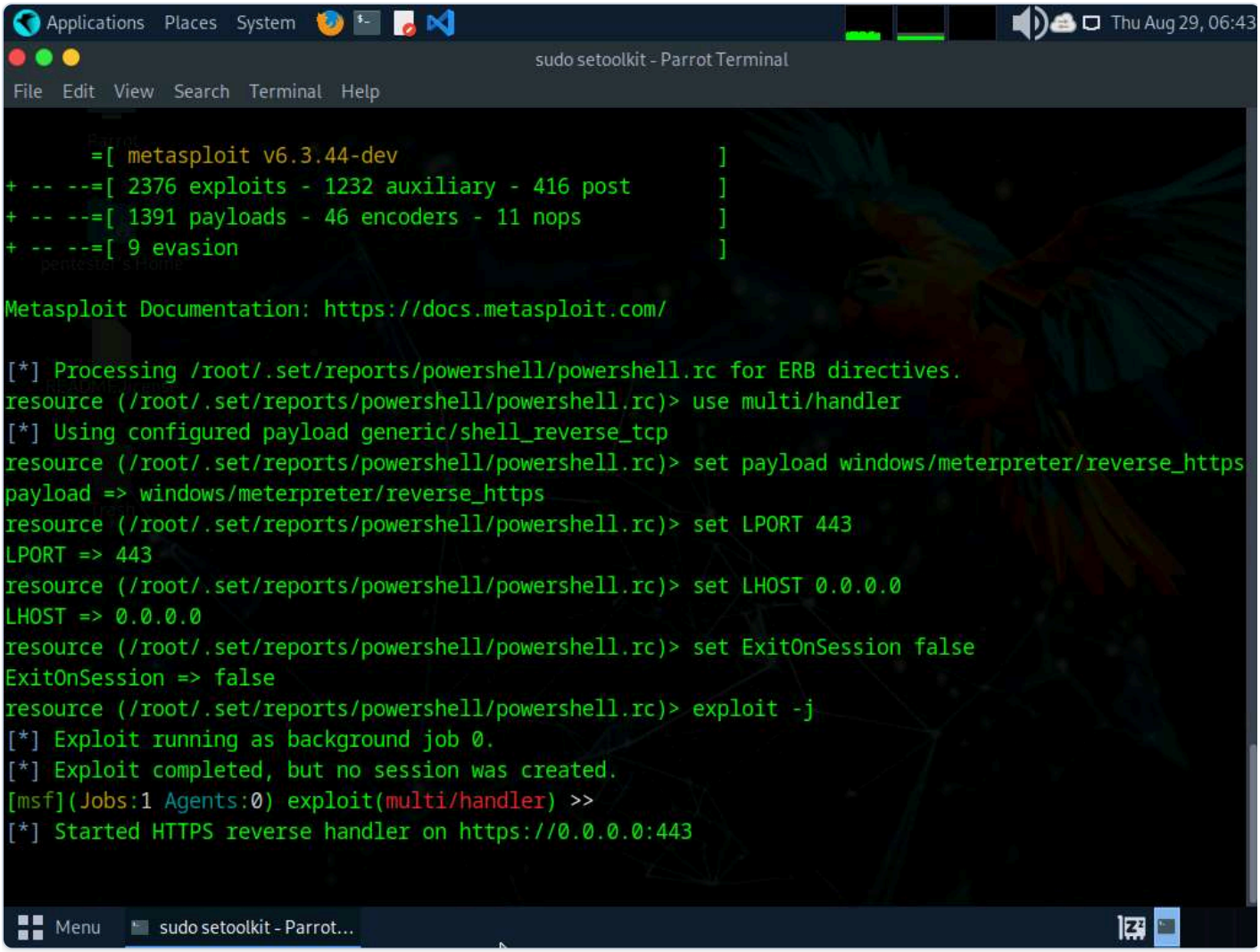
1. Enter custom reverse connection port:
   443



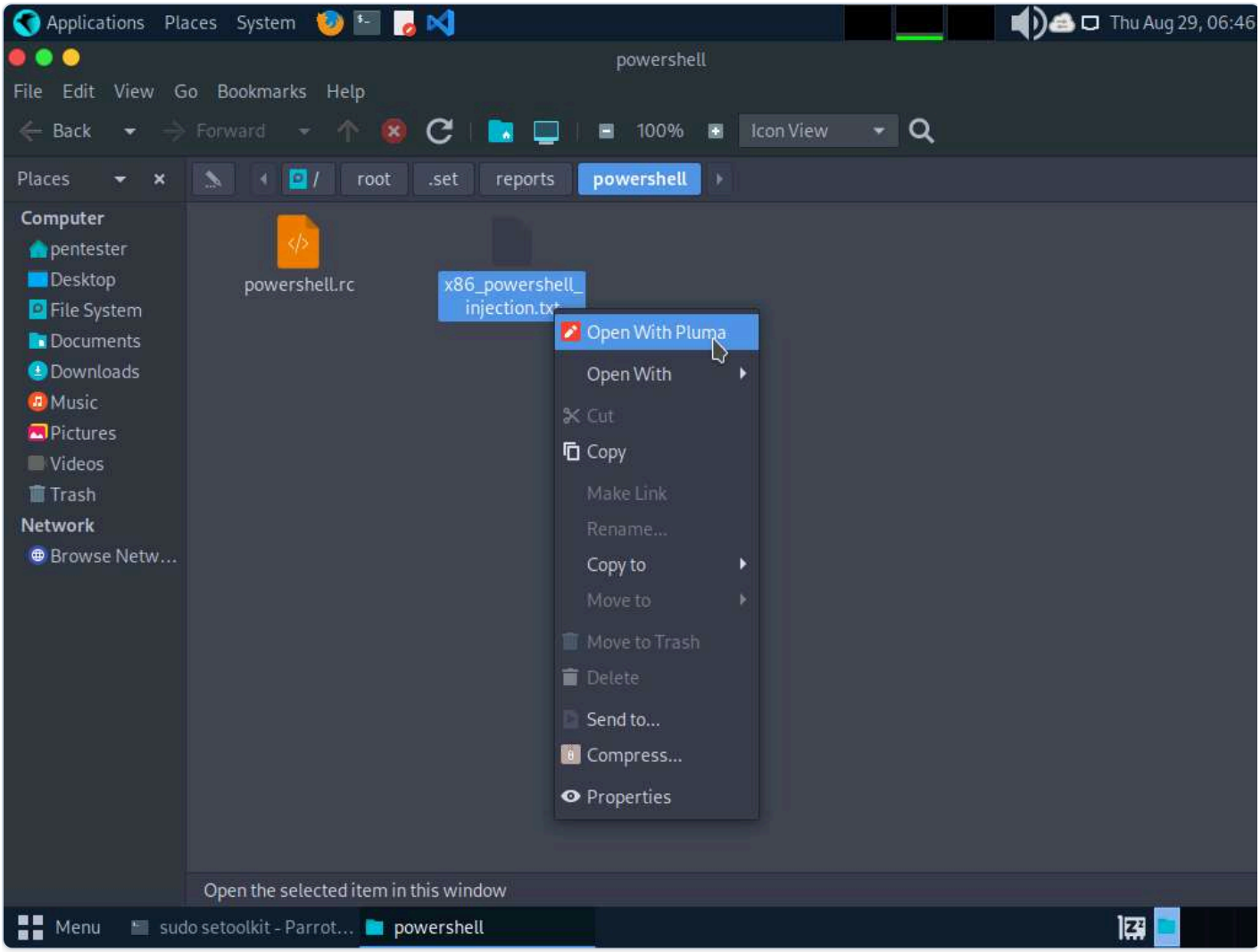
1. When prompted *Do you want to start the payload handler now?*, type **yes** and press Enter.

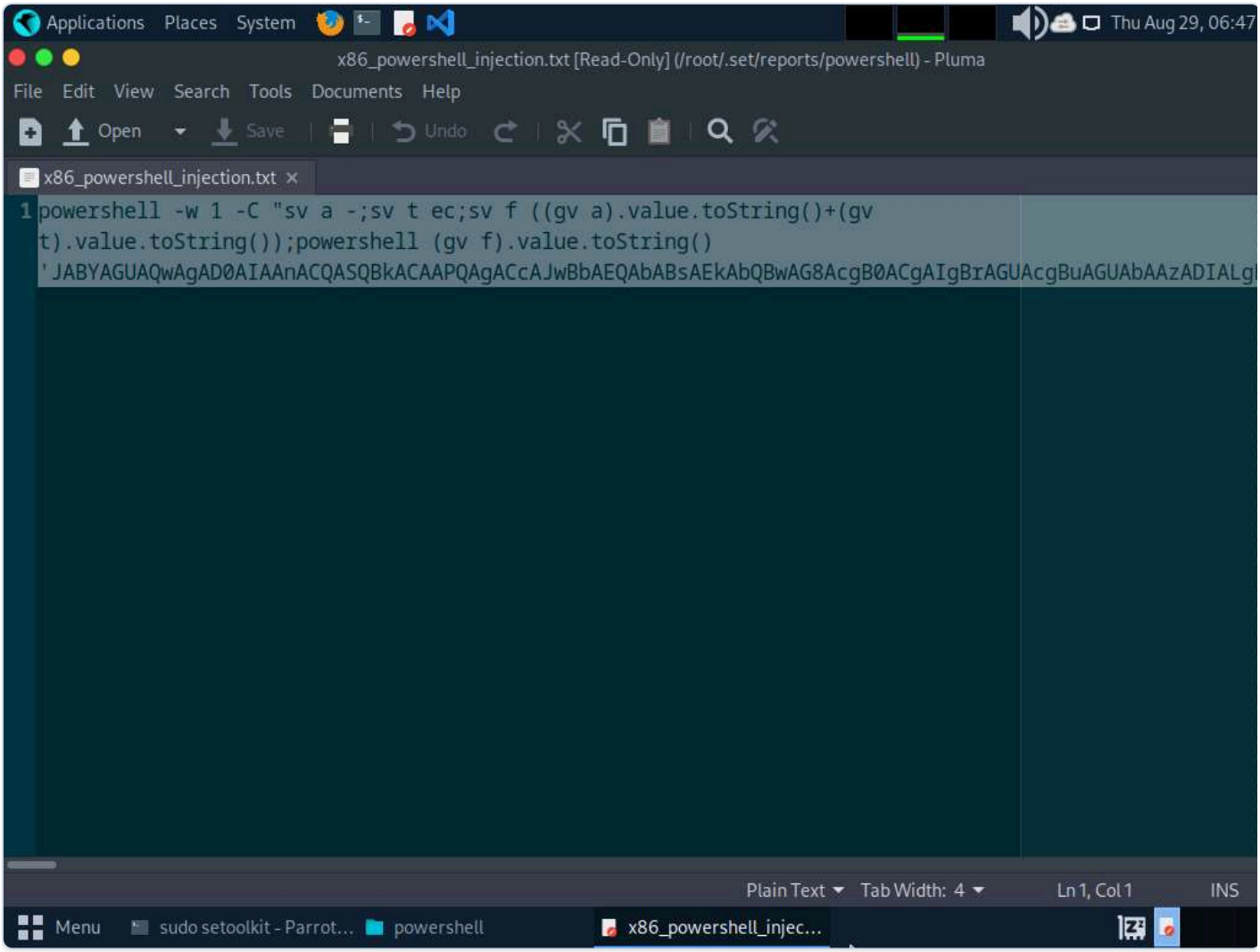


1. SET initializes the Metasploit multi-handler in the background and saves the generated payload to: **/root/.set/reports/powershell/x86_powershell_injection.txt**

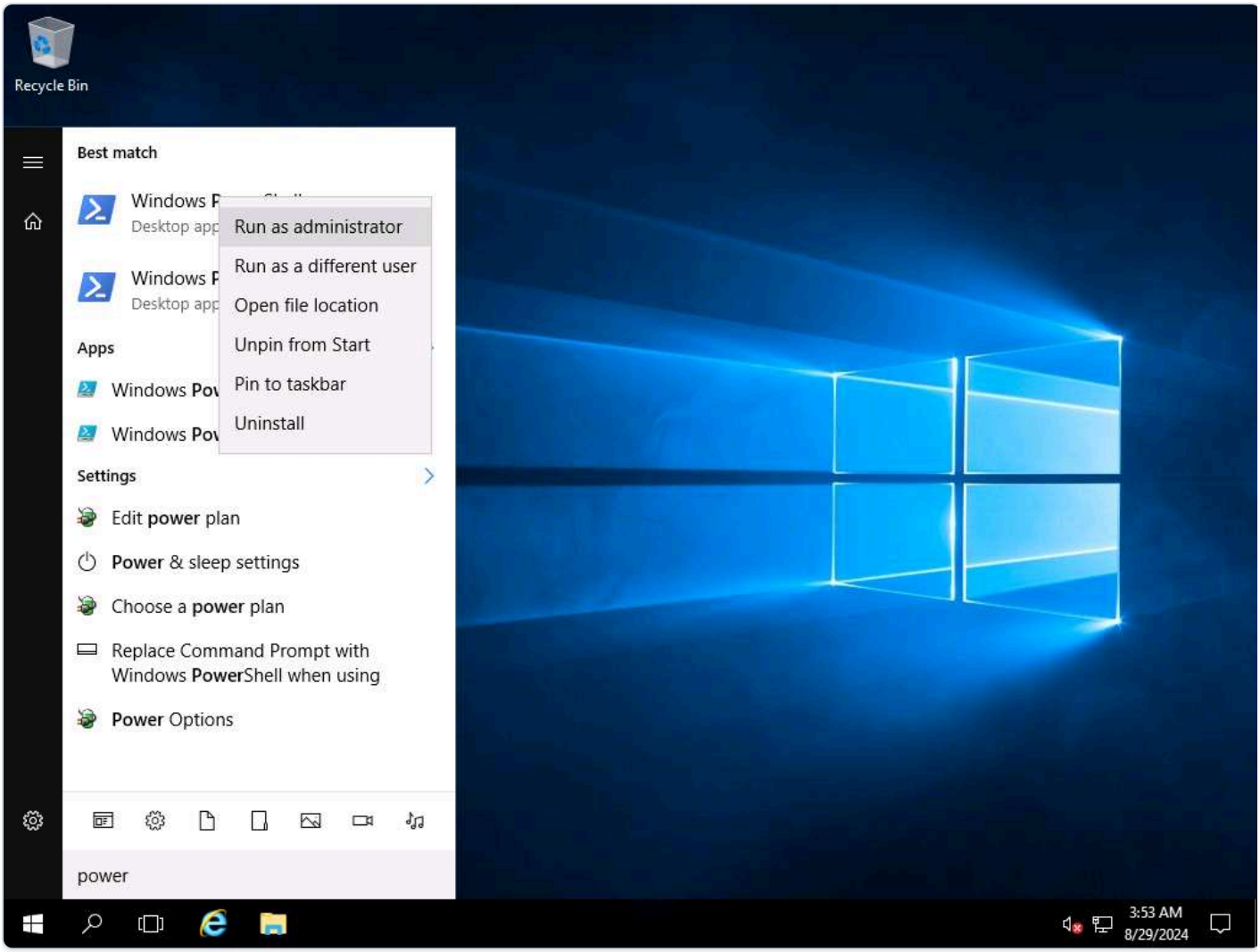


1. Open a file manager window on Parrot Security, press **Ctrl+H** to reveal hidden folders, and open `/root/.set/reports/powershell/x86_powershell_injection.txt`. Copy the full PowerShell payload text.

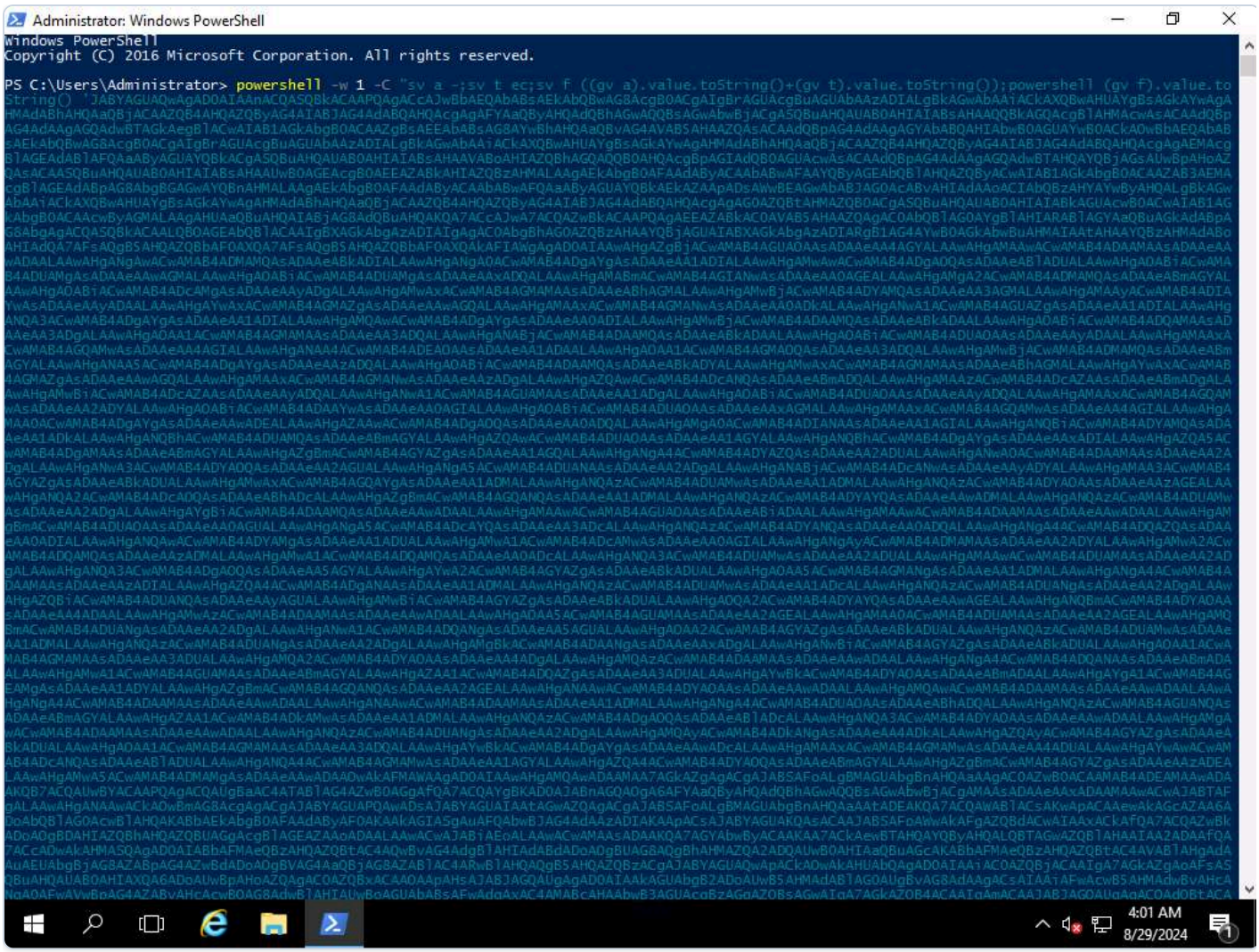

Applications
Places
System
Thu Aug 29, 06:46
powershell
File
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Go
Bookmarks
Help
Back
Forward
100%
Icon View
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/
root
.set
reports
powershell
Computer
pentester
Desktop
File System
Documents
Downloads
Music
Pictures
Videos
Trash
Network
Browse Netw...
powershell.rc
x86_powershell_injection.txt
Open With Pluma
Open With
Cut
Copy
Make Link
Rename...
Copy to
Move to
Move to Trash
Delete
Send to...
Compress...
Properties
Open the selected item in this window
Menu
sudo setoolkit - Parrot...
powershell



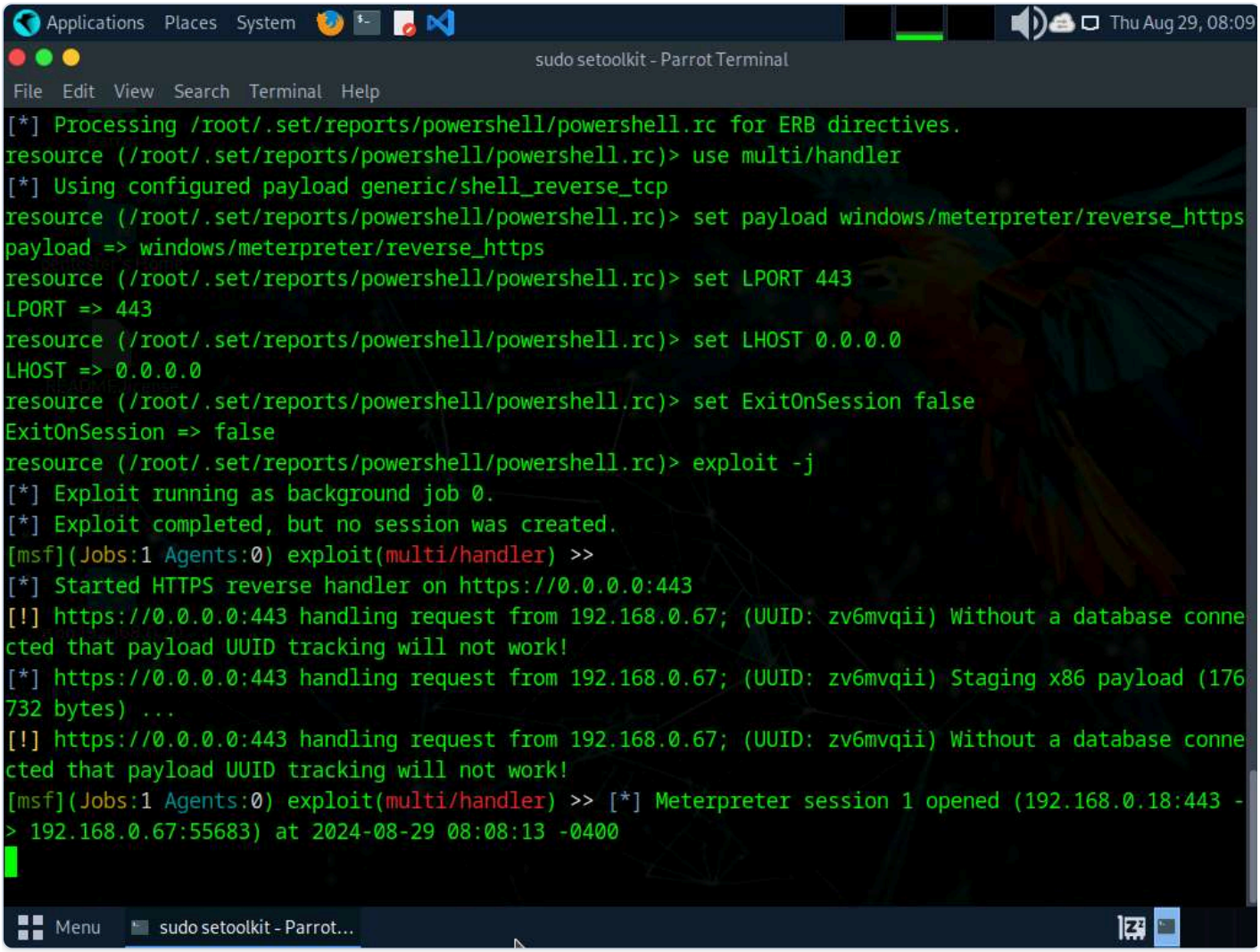
1. Switch to **Windows Server 2019** and log in with password **Pa$$w0rd**.
2. Search for **PowerShell**, right-click **Windows PowerShell**, and select **Run as administrator**.



1. Paste the copied SET PowerShell payload script into the Windows PowerShell window and press Enter.



1. Observe that the PowerShell window closes immediately upon payload execution. Switch to Parrot Security and verify that an active **Meterpreter** session has successfully opened.



1. Close all active terminal sessions and handlers.

This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 7.4.1.1** On the Parrot Security machine, utilize the Social-Engineer Toolkit (SET) to generate a PowerShell script. Transfer the script to the Windows Server 2019 system and execute it in Windows PowerShell to establish a Meterpreter session on the Parrot Security machine. Enter the serial number of the PowerShell Attack Vector under the Social Engineering Attacks module. [!SUCCESS] ✅ **Answer:** **9**